1.10 Integration

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs c20–c33 (Calculus)

What this topic covers. Integration as the reverse of differentiation and as area, the standard integrals, the three techniques — substitution, by parts and partial fractions — and definite integration for areas between curves.

1. Integration as the reverse of differentiation

$$\int x^n \, \mathrm{d}x = \frac{x^{n+1}}{n+1} + c \qquad (n \neq -1)$$

The **constant of integration** c is essential for an indefinite integral: differentiating any constant gives zero, so the original constant cannot be recovered.

The excluded case $n = -1$ is covered separately:

$$\int \frac{1}{x} \, \mathrm{d}x = \ln|x| + c$$

The modulus in $\ln|x|$ matters — it allows the result to apply for negative x as well. Dropping it is a standard deduction.

2. Standard integrals

$f(x)$	$\int f(x)\,\mathrm{d}x$
$x^n \ (n \neq -1)$	$\dfrac{x^{n+1}}{n+1} + c$

$\dfrac{1}{x}$	$\ln\lvert x\rvert + c$
e^x	$\mathrm{e}^x + c$
e^{kx}	$\dfrac{\mathrm{e}^{kx}}{k} + c$
$\cos x$	$\sin x + c$
$\sin x$	$-\cos x + c$
$\sec^2 x$	$\tan x + c$
$\tan x$	$\ln\lvert\sec x\rvert + c$

Note the sign asymmetry: differentiating $\cos$ introduces a minus, and so does integrating $\sin$. $\int \sin x \, \mathrm{d}x = -\cos x + c$ is the one people write without the minus.

3. Definite integration and area

The **Fundamental Theorem of Calculus** connects the two views of integration:

$$\int_a^b f(x)\,\mathrm{d}x = \big[F(x)\big]_a^b = F(b) - F(a)$$

where $F'(x) = f(x)$.

No constant of integration is needed for a definite integral — it cancels in the subtraction.

Worked example. Evaluate $\displaystyle\int_1^3 (2x+1)\,\mathrm{d}x$.

$$\big[x^2 + x\big]_1^3 = (9+3) - (1+1) = 10$$

A region **below** the x-axis contributes a negative amount. If a curve crosses the axis within the limits, split the integral at the crossing and add the *magnitudes*, or the positive and negative parts will cancel and understate the area.

Area between two curves

$$\text{Area} = \int_a^b \left(y_{\text{upper}} - y_{\text{lower}}\right) \mathrm{d}x$$

Worked example. Find the area enclosed between $y = x^2$ and $y = 2x$.

They meet where $x^2 = 2x$, i.e. $x = 0$ and $x = 2$. Between these the line is above the parabola:

$$\int_0^2 (2x - x^2)\,\mathrm{d}x = \left[x^2 - \frac{x^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$$

Worked example. Find the area between $y = 4 - x^2$ and the x-axis, between its roots.

Roots at $x = \pm 2$:

$$\int_{-2}^2 (4 - x^2)\,\mathrm{d}x = \left[4x - \frac{x^3}{3}\right]_{-2}^2 = \left(8 - \tfrac{8}{3}\right) - \left(-8 + \tfrac{8}{3}\right) = \frac{32}{3}$$

4. Integration by substitution

Reverse of the chain rule. Choose u so that $\mathrm{d}u$ absorbs the remaining factor.

Worked example. Find $\displaystyle\int 2x\,\mathrm{e}^{x^2}\,\mathrm{d}x$.

Let $u = x^2$, so $\mathrm{d}u = 2x\,\mathrm{d}x$:

$$\int \mathrm{e}^u\,\mathrm{d}u = \mathrm{e}^u + c = \mathrm{e}^{x^2} + c$$

Worked example. Find $\displaystyle\int (2x+1)^5\,\mathrm{d}x$.

Let $u = 2x+1$, so $\mathrm{d}u = 2\,\mathrm{d}x$:

$$\tfrac{1}{2}\int u^5\,\mathrm{d}u = \frac{u^6}{12} + c = \frac{(2x+1)^6}{12} + c$$

Worked example. Find $\displaystyle\int \frac{1}{x\ln x}\,\mathrm{d}x$.

Let $u = \ln x$, so $\mathrm{d}u = \frac{1}{x}\,\mathrm{d}x$:

$$\int \frac{1}{u}\,\mathrm{d}u = \ln|u| + c = \ln|\ln x| + c$$

For a **definite** integral, either change the limits to match u, or substitute back to x before applying the original limits. Applying the x-limits to a u-expression is a serious and common error.

The standard pattern

$$\int \frac{f'(x)}{f(x)}\,\mathrm{d}x = \ln|f(x)| + c$$

Worked example. $\displaystyle\int_0^1 \frac{2x}{x^2+1}\,\mathrm{d}x = \Big[\ln(x^2+1)\Big]_0^1 = \ln 2$

5. Integration by parts

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

Reverse of the product rule. Choose u to be the part that **simplifies** when differentiated.

Worked example. Find $\displaystyle\int x\mathrm{e}^x\,\mathrm{d}x$.

Take $u = x$ (differentiates to 1) and $\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^x$:

$$xe^x - \int e^x \, dx = xe^x - e^x + c = e^x(x-1) + c$$

Worked example. Find $\int \ln x \, dx$.

Write the integrand as $1 \times \ln x$, and take $u = \ln x$, $\frac{dv}{dx} = 1$:

$$x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - x + c$$

Worked example. Find $\int x^2 e^x \, dx$.

Parts once gives $x^2 e^x - \int 2xe^x \, dx$. Applying parts a **second** time to the remaining integral:

$$x^2 e^x - (2xe^x - 2e^x) + c = e^x \left(x^2 - 2x + 2\right) + c$$

For $\ln x$, the trick is to treat it as $1 \times \ln x$ — there is no other way to start, since $\ln x$ has no elementary antiderivative on its own. For $\int e^x \sin x \, dx$, applying parts twice returns the original integral, which you then solve for algebraically.

6. Integration using partial fractions

Split the fraction first, then integrate each simple term.

Worked example. Find $\int \frac{2x+3}{(x+1)(x+2)} \, dx$.

In partial fractions this is $\frac{1}{x+1} + \frac{1}{x+2}$, so the integral is

$$\ln|x+1| + \ln|x+2| + c$$

7. Trigonometric integrals via identities

Powers of sine and cosine are handled by converting them with a double angle identity.

Worked example. Find $\int \cos^2 x \, \mathrm{d}x$.

Use $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$:

$$\tfrac{1}{2}\int (1 + \cos 2x)\, \mathrm{d}x = \tfrac{1}{2}\left(x + \tfrac{\sin 2x}{2}\right) + c$$

Worked example. Evaluate $\int_0^{\pi/2} \sin x \cos x \, \mathrm{d}x$.

Since $\sin x \cos x = \frac{1}{2}\sin 2x$:

$$\tfrac{1}{2}\int_0^{\pi/2} \sin 2x \, \mathrm{d}x = \tfrac{1}{2}\left[-\tfrac{\cos 2x}{2}\right]_0^{\pi/2} = \tfrac{1}{4}(1 + 1) = \tfrac{1}{2}$$

8. Choosing a method

Integrand looks like	Method
A power, or a standard form	Direct
Inner function with its derivative alongside	Substitution
$\dfrac{f'(x)}{f(x)}$	Straight to $\ln\lvert f(x)\rvert$
Product of unlike functions, e.g. $x\mathrm{e}^x$, $x \sin x$, $\ln x$	By parts
Algebraic fraction with a factorisable denominator	Partial fractions
Even powers of $\sin$ or $\cos$	Double angle identity

9. Common pitfalls

- Omitting $+c$ on an indefinite integral.
- Dropping the modulus in $\ln|x|$.
- Applying $\dfrac{x^{n+1}}{n+1}$ when $n = -1$.
- Letting positive and negative areas cancel when a curve crosses the axis.
- Applying x-limits to a u-expression after substituting.
- Choosing u in parts so that it becomes *more* complicated on differentiating.
- Integrating $\sin x$ to $\cos x$ without the minus.
- Subtracting the curves the wrong way round for an area, giving a negative answer.